Aggregating Distortions in Networks with Multi-Product Firms

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Motivation

- Aggregate productivity determined by technological and allocative efficiency
- Large literature decomposes aggregate TFP into resource misallocation and a technological residual
- Standard assumption is that each firm produces a single product
- In the data, most transactions involve multiproduct firms (e.g., 99% in Chile)

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- With multiproduct firms, the allocation of resources across products within firms matters
- Assigning inputs to outputs within firms is hard or even impossible (e.g., oil refinery)

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- In the data, most transactions involve multiproduct firms (e.g., 99% in Chile)
- With multiproduct firms, the allocation of resources across products within firms matters
- Assigning inputs to outputs within firms is hard or even impossible (e.g., oil refinery)

How do multiproduct firms with joint (non-separable) production technology change the measurement of misallocation?

This Paper

Framework for measuring allocative efficiency with multi-product firms and joint production.

- Joint production captures how easy it is for firms to adjust their product mix (e.g., gas/diesel)
- Finding: technological barriers in adjusting product mix attenuate reallocation effects on TFP
- Develop sufficient stats using product level prices to capture curvature of firm-level PPF

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- Joint production captures how easy it is for firms to adjust their product mix (e.g., gas/diesel)
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- Develop sufficient stats using product level prices to capture curvature of firm-level PPF

Quantify the framework using firm–product-level transactions and prices in Chile

- Conduct aggregate growth accounting to decompose observed TFP changes into allocative and technical efficiency in Chile, where aggregate TFP declined during 2016–2022.
- Estimate the curvature of firms' PPFs and use the parametric model to conduct counterfactuals.
- Ignoring joint production overstates the reallocation effects on aggregate TFP.

Related Papers

- Multi-product firms & Joint production
  - Multi-product firms: Klette and Kortum (2004); Bernard et al. (2010, 2011); Mayer et al. (2014); De Loecker et al. (2016); Hottman et al. (2016); Mayer et al. (2021); Wang and Yang (2023)
  - Joint production: Diewert (1971); Lau (1972); Hall (1973); Dhyne et al. (2017); Almunia et al. (2021); Dhyne et al. (2022); Ding (2023); Carrillo et al. (2023); Cairncross et al. (2025)
  - Study misallocation with multi-product firms that engage in joint production
  - Provide empirical evidence consistent with joint production
- Aggregation & (In)efficiency in production networks
  - Hulten (1978); Basu and Fernald (2002); Restuccia and Rogerson (2008); Petrin and Levinsohn (2012); Jones (2013); Liu (2019); Baqaee and Farhi (2020); Bigio and La'O (2020); Kikkawa (2022); Baqaee et al. (2023)
  - Generalize theory to allow for joint production and quantify using firm-to-firm product level data

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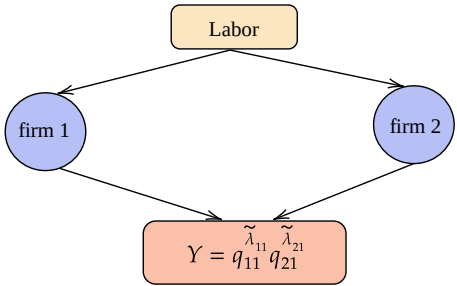
3 Quantifying Misallocation

4 Parametric Counterfactual

Illustrative Example

- Demonstrate that aggregate TFP can change without technological progress
- Illustrate how the change in TFP varies with and without joint production
- Show how sufficient statistics work
- Much of the intuition from the simple example survives in general setup

Illustrative Example with Single Product Firms

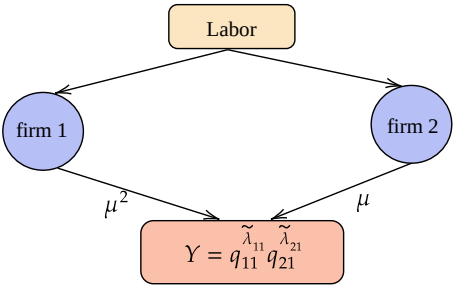


- First index: firm. Second index: each firm's product
- Production functions: $q_{11} = L_{11}, \;\; q_{21} = L_{21}$
- Each firm sets its price as $p_{i1} = \mu_{i1} mc_{i1}$, where μ_{i1} is exogenous wedge (markup)
- Household maximizes $Y = q_{11}^{\tilde{\lambda}_{11}} q_{21}^{\tilde{\lambda}_{21}}$ subject to

$$\sum_{i=1}^{2} p_{i1} q_{i1} = wL + \sum_{i=1}^{2} (1 - 1/\mu_{i1}) p_{i1} q_{i1},$$

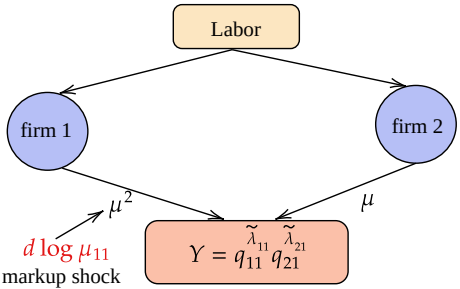
- Market clearing condition: $L = L_{11} + L_{21}$
- $TFP = Y/L$

Misallocation with Single Product Firms



- Firm 1 is underproduced due to higher wedge

Misallocation with Single Product Firms



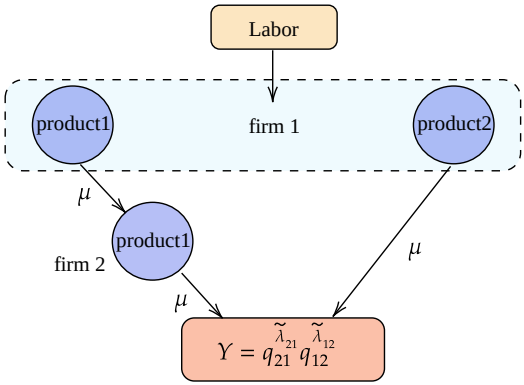
- First order TFP response to markup change:

$$\Delta \log TFP = \tilde{\lambda}_{11} \underbrace{\left(\frac{\bar{\mu}}{\mu_{11}} - 1 \right)}_{<0} d \log \mu_{11}$$

where $\bar{\mu} = \left(\tilde{\lambda}_{11} \mu_{11}^{-1} + \tilde{\lambda}_{21} \mu_{21}^{-1} \right)^{-1}$

- An increase in wedges for products with higher wedges decreases TFP

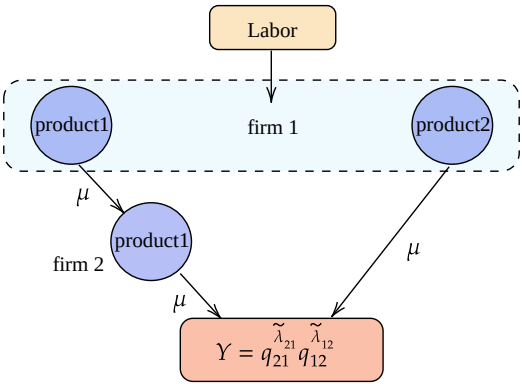
Now with Multi-Product Firms but not Joint Production



- Introduce multi-product firms and downstream firms:
- Production functions:
  $q_{11} = L_{11}, \;\; q_{12} = L_{12}, \;\; q_{21} = q_{11}$
- Household: $Y = q_{21}^{\tilde{\lambda}_{21}} q_{12}^{\tilde{\lambda}_{12}}$
- Market clearing:

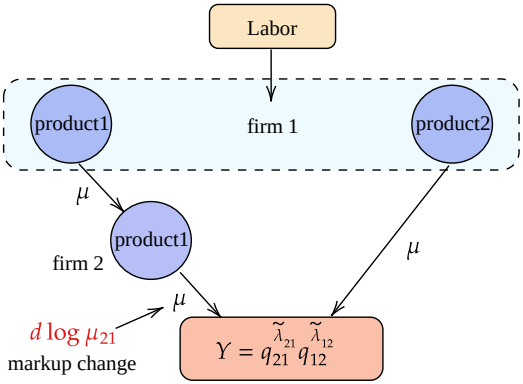
$$L = L_{11} + L_{12}$$

Misallocation with Multi-Product Firm



- Product 1 is underproduced due to downstream wedge
- Define $\Gamma_{11} = \mu^2$ and $\Gamma_{12} = \mu$ to be cumulative wedges
- Define $\tilde{\lambda}_{11} = \tilde{\lambda}_{21}$ to capture the (indirect) cost effect of firm 1's product 1 on final demand

Misallocation with Multi-Product Firm



- Cumulative wedges: $\Gamma_{11} = \mu^2$ and $\Gamma_{12} = \mu$
- First order TFP response to markup change:

$$\Delta \log TFP = \tilde{\lambda}_{21} \underbrace{\left(\frac{\bar{\Gamma}_1}{\Gamma_{11}} - 1 \right)}_{<0} d \log \mu_{21}$$

where $\bar{\Gamma}_1 = \left(\tilde{\lambda}_{11} \Gamma_{11}^{-1} + \tilde{\lambda}_{12} \Gamma_{12}^{-1} \right)^{-1}$

- An increase in cumulative wedges for products with high cumulative wedges decreases TFP

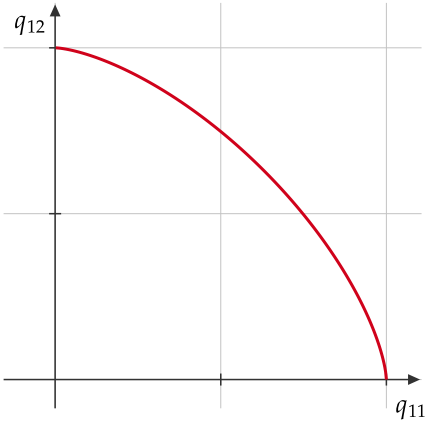
Production Possibility Frontier with Joint Production

Multiproduct firms use shared input to make multiple outputs
(Diewert (1971); Lau (1972); Hall (1973)) details

- Constant-Elasticity of Transformation (CET)

$$\underbrace{\left(q_{11}^{\frac{\sigma+1}{\sigma}} + q_{12}^{\frac{\sigma+1}{\sigma}}\right)^{\frac{\sigma}{\sigma+1}}}_{\text{output bundle}} = L$$

- σ is elasticity of transformation



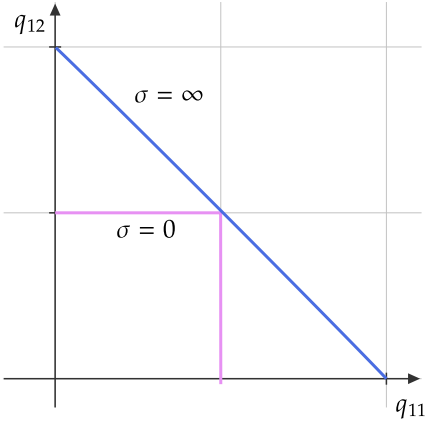
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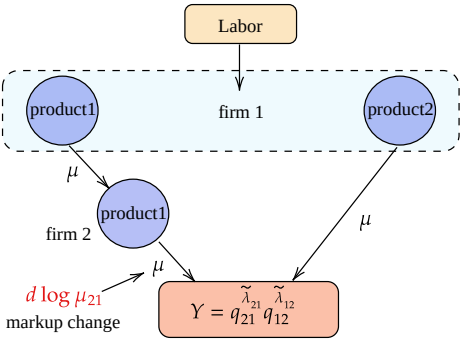
- Constant-Elasticity of Transformation (CET)

$$\underbrace{\left(q_{11}^{\frac{\sigma+1}{\sigma}} + q_{12}^{\frac{\sigma+1}{\sigma}}\right)^{\frac{\sigma}{\sigma+1}}}_{\text{output bundle}} = L$$

- $\sigma \to \infty$: separable production
- $\sigma \to 0$: outputs can only be produced in a constant ratio



Misallocation with Joint Production



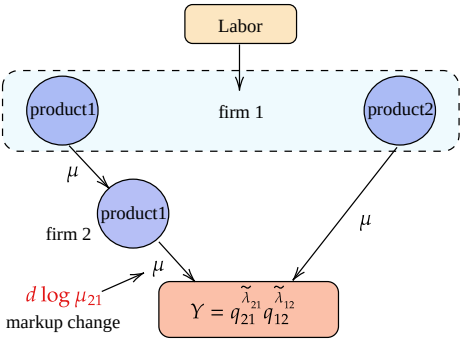
$$\left(q_{11}^{\frac{\sigma+1}{\sigma}} + q_{12}^{\frac{\sigma+1}{\sigma}} \right)^{\frac{\sigma}{\sigma+1}} = L$$

- First order TFP response to the same markup change:

$$\Delta \log TFP = \left(1 - \frac{1}{\sigma+1}\right) \tilde{\lambda}_{21} \left(\frac{\bar{\Gamma}_1}{\Gamma_{11}} - 1 \right) d \log \mu_{21}$$

depends on ease of adjusting product mix, σ

Misallocation with Joint Production



$$\left(q_{11}^{\frac{\sigma+1}{\sigma}} + q_{12}^{\frac{\sigma+1}{\sigma}} \right)^{\frac{\sigma}{\sigma+1}} = L$$

- First order TFP response to the same markup change:

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depends on ease of adjusting product mix, σ

$$\Delta \log TFP = \begin{cases} \tilde{\lambda}_{21} \left(\frac{\bar{\Gamma}_1}{\bar{\Gamma}_{11}} - 1 \right) d \log \mu_{21} & \text{if } \sigma \to \infty \\ 0 & \text{if } \sigma \to 0 \end{cases}$$

- Joint production attenuates TFP response

Using Prices Instead of PPF Curvature

- To derive sufficient statistics, rewrite the TFP response in a way that is not dependent on σ
- With joint production, relative price changes are associated with changes in production ratio:

$$d\log(p_{11}/p_{12}) = \frac{1}{\sigma} d\log(q_{11}/q_{12})$$

Using Prices Instead of PPF Curvature

- With joint production, relative price changes are associated with changes in production ratio:

$$d\log(p_{11}/p_{12}) = \frac{1}{\sigma} d\log(q_{11}/q_{12})$$

- Solving for prices: $d\log(p_{11}/p_{12}) = -\frac{1}{\sigma+1} d\log\mu_{21}$, the TFP response is expressed by: deri

$$\begin{aligned}\Delta\log TFP &= \left(1 - \frac{1}{\sigma+1}\right)\tilde{\lambda}_{21}\left(\frac{\bar{\Gamma}_1}{\Gamma_{11}} - 1\right) d\log\mu_{21} \\ &= \underbrace{\tilde{\lambda}_{21}\left(\frac{\bar{\Gamma}_1}{\Gamma_{11}} - 1\right) d\log\mu_{21}}_{\text{Single-Product Term}} + \underbrace{d\log(p_{11}/p_{12})\,\tilde{\lambda}_{21}\left(\frac{\bar{\Gamma}_1}{\Gamma_{11}} - 1\right)}_{\text{Multi-Product Term}}\end{aligned}$$

Using Prices Instead of PPF Curvature

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$$\Delta\log TFP = \left(1 - \frac{1}{\sigma+1}\right)\tilde{\lambda}_{21}\left(\frac{\bar{\Gamma}_1}{\Gamma_{11}} - 1\right) d\log\mu_{21}$$

$$= \underbrace{\underbrace{-\underbrace{d\log\Lambda}_{\text{Labor Share}} - \tilde{\lambda}_{21} d\log\mu_{21}}_{\text{Single-Product Term}} + \underbrace{Cov_{\boldsymbol{s}_1}\left(d\log p_{(1,\cdot)}, \frac{\bar{\Gamma}_1}{\Gamma_{(1,\cdot)}}\right)}_{\text{Multi-Product Term}}}$$

- If $\sigma \to \infty$, multi-product term is zero
- If $\sigma \to 0$, $\Delta\log TFP$ is zero

Sufficient Statistics for Simple Example

Proposition

In the simple example, TFP response to the markup shock can be expressed as

$$\Delta \log TFP = \underbrace{Cov_{\boldsymbol{s}_1}\left(d\log p_{(1,\cdot)}, \frac{\bar{\Gamma}_1}{\Gamma_{(1,\cdot)}}\right)}_{\text{Multi-Product Term}} \underbrace{- \underbrace{d\log\Lambda}_{\text{Labor Share}} - \tilde{\lambda}_{21} d\log\mu_{21}}_{\text{Single-Product Term}}$$

where $d\log p_{(1,\cdot)} = (d\log p_{11}, d\log p_{12})$, $\Gamma_{(1,\cdot)} = (\Gamma_{11}, \Gamma_{12})$, $\bar{\Gamma}_1 = \left(\tilde{\lambda}_{11}\Gamma_{11}^{-1} + \tilde{\lambda}_{12}\Gamma_{12}^{-1}\right)^{-1}$ *and* $\boldsymbol{s}_1 = \left(\tilde{\lambda}_{11}, \tilde{\lambda}_{12}\right)$

taste shock

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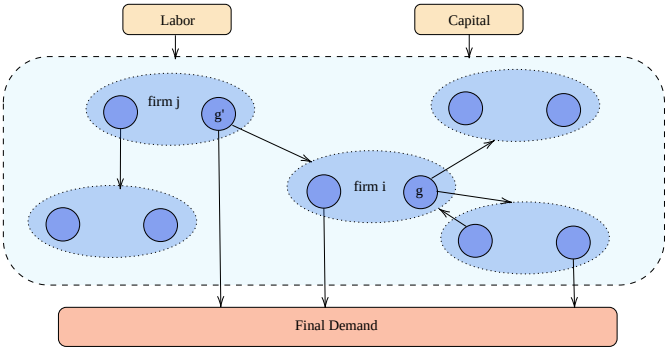
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General Theory for Measurement

- Introduce a general framework to derive sufficient statistics that can accommodate any number of products and arbitrary input-output linkages



General Setup

- Firm $i \in \mathcal{N}$ makes product $g \in \mathcal{G}$ using g' from firm j and factors as inputs
- CRS joint production with firm-level technology shock (technology shock can be product specific with additional assumption): details

$$F_i^Q\left(\underbrace{\{q_{ig}\}_{g\in\mathcal{G}}}_{\text{outputs}}\right) = A_i F_i^X\left(\underbrace{\{x_{i,jg'}\}_{j\in\mathcal{N},g'\in\mathcal{G}}}_{\text{product } g' \text{ from j}}, L_i, K_i\right)$$

- Firm i sets its products' price as $p_{ig} = \mu_{ig} mc_{ig}$, where μ_{ig} is the wedge (markup)
- Agnostic about microfoundations of wedges (e.g., market power, sticky prices, etc.)

General Setup

- Final demand as maximizer of homothetic aggregator, $Y = D(c_{ig}, ..., c_{NG})$: subject to:

$$\underbrace{\sum_{i\in\mathcal{N}}\sum_{g\in\mathcal{G}} p_{ig}c_{ig}}_{\text{Final goods expenditure}} = \underbrace{\sum_{f\in\{L,K\}} w_f L_f}_{\text{Factor income}} + \underbrace{\sum_{i\in\mathcal{N}}\sum_{g\in\mathcal{G}} (1-1/\mu_{ig})\, p_{ig}q_{ig}}_{\text{Profits}}$$

- Resource constraints: $q_{ig} = c_{ig} + \sum_{j\in\mathcal{N}} x_{jig}$, $\sum_{i\in\mathcal{N}} L_i = L$, $\sum_{i\in\mathcal{N}} K_i = K$

Definition (General Equilibrium)

Given productivities $\boldsymbol{A}$ *and product-level markups* $\boldsymbol{\mu}$*, an equilibrium consists of prices* p_{ig}, w_f*, intermediate input choices* $x_{ijg'}$*, factor inputs* (L_i, K_i)*, outputs* q_{ig}*, and consumption* c_{ig}*, such that (i) each firm minimizes its costs and sets prices as markups over marginal costs, (ii) final demand maximizes the aggregator subject to budget constraints, (iii) and all goods and factor markets clear.*

Growth Accounting with Multi-Product Firms

Proposition

To the first order, aggregate TFP is decomposed into allocative efficiency and technology :

$$d\log TFP = \underbrace{\underbrace{\sum_i \tilde{\lambda}_i Cov_{\boldsymbol{s}_i}\left(d\log p_{(i,\cdot)}, \frac{\bar{\Gamma}_i}{\Gamma_{(i,\cdot)}}\right)}_{\text{Multi-Product Term}} \underbrace{-\sum_f \tilde{\Lambda}_f d\log \Lambda_f - \sum_i \tilde{\lambda}_i d\log \mu_i}_{\text{Single-Product Term}}}_{\Delta \text{Allocative Efficiency}} + \underbrace{\sum_i \tilde{\lambda}_i d\log A_i}_{\Delta \text{Technology(residual)}}$$

where $d\log p_{(i,\cdot)} = (d\log p_{i1}, ..., d\log p_{iG})$, $\Gamma_{(i,\cdot)} = (\Gamma_{i1}, ..., \Gamma_{iG})$ *and* $\bar{\Gamma}_i = \mathbb{E}_{\boldsymbol{s}_i}\left(\Gamma^{-1}_{(i,g)}\right)^{-1}$

Wedges and Domar weights depend on markups, sales, and expenditure shares.

IO matrix | cumulative wedge | formal definition | efficient benchmark | interpretation

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Quantifying Misallocation

- Perform growth accounting to decompose Chile's cumulative TFP changes into allocative efficiency and technology components
- Construct sufficient statistics using firm-to-firm transaction data from Chile
- Compare results with and without accounting for joint production

Data: Administrative Chilean Invoice between Firms

- Electronic invoices between all formal firms, including service firms
- Information on buyers, sellers, product, quantity, and price distribution
- Combined with balance sheets information

Figure: Example of Invoice



Construction of Sufficient Statistics

- Domar weights: from firm-to-firm transactions and balance sheets details
- Cumulative wedge:

$$\Gamma_{ig} \equiv \underbrace{\frac{\tilde{\lambda}_{ig}}{\text{sales share}_{ig}}}_{\text{downstream wedge}} \times \underbrace{\mu_{ig}}_{\text{own markup}}$$

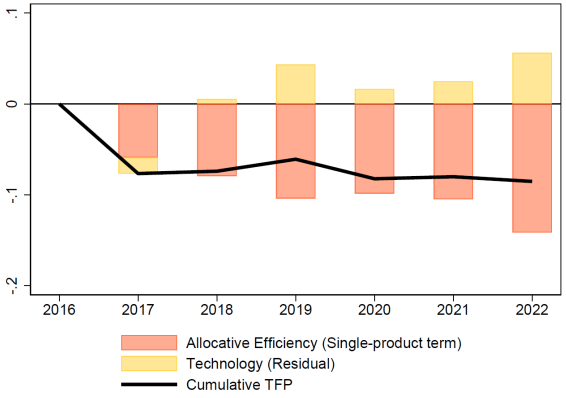
  - Baseline: estimated product-level markups, adapting Dhyne et al. (2022) to CET production with estimated $\sigma = 1.589$ markup estimation elasticity estimate
  - Robustness: replace own markups in the multi-product term with firm-level sales/cost ratios accounting markups

- Factor shares and aggregate TFP: aggregated microdata wedge variation

Cumulative TFP Decomposition: Ignoring Multi-Product Term

$$\Delta \text{TFP} = \underbrace{\Delta \text{Technology}}_{\text{Residual}} + \underbrace{\Delta \text{Single-Product Term}}_{\text{Allocative Efficiency}}$$

- Large negative allocative efficiency (-14.1%): firms with high cumulative wedges shrink and firms with low cumulative wedges expand in terms of input use
- Since TFP does not fall by as much, the technology residual must be large and positive (+5.6%)

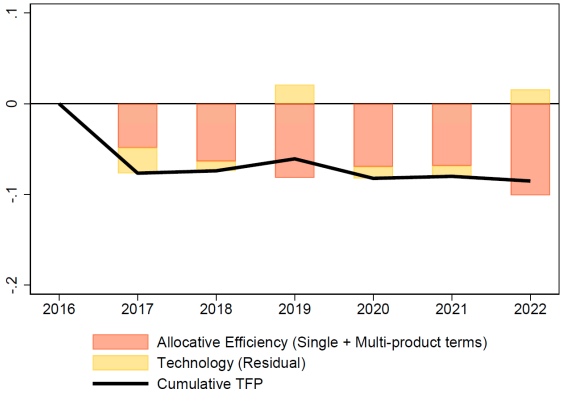


Cumulative TFP Decomposition with Multi-Product Term

$$\Delta \text{TFP} = \underbrace{\Delta \text{Technology}}_{\text{Residual}} + \underbrace{\Delta \text{Single-Product Term} + \Delta \text{Multi-Product Term}}_{\text{Allocative Efficiency}}$$

- Adding the multi-product term attenuates the strong reallocation effects (14.1% $\rightarrow$ 10.1%; 40% overstatement if ignored)
- Technology residual is now small (+1.6%)

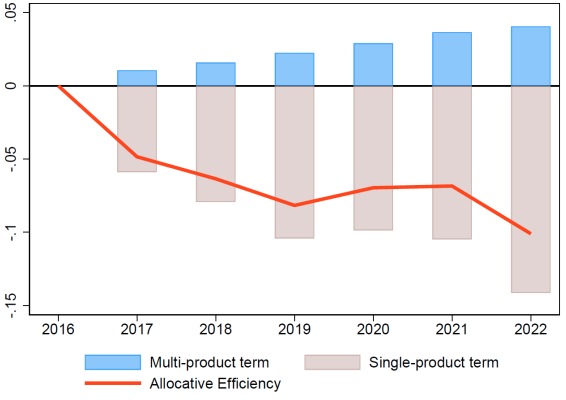
accounting-markup robustness



Cumulative TFP Decomposition with Multi-Product Term

$$\Delta \text{TFP} = \underbrace{\Delta \text{Technology}}_{\text{Residual}} + \underbrace{\Delta \text{Single-Product Term} + \Delta \text{Multi-Product Term}}_{\text{Allocative Efficiency}}$$

- The multi-product term attenuates the single-product term as the theory suggests (offset: 4.0 percentage points)
- Rising relative prices limit the expansion of low cumulative wedge products, mitigating negative reallocation



accounting-markup robustness

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Quantifying Distance to Frontier

- Use parametric model to measure TFP gains (distance to frontier) when removing all wedges:

$$\text{Distance to Frontier} \equiv \log \frac{Y(1)}{Y(\boldsymbol{\mu})}$$

- Parameterizing the general framework's production technologies using CET and CES specifications:

$$F_i^Q\left(\underbrace{\{q_{ig}\}_{g\in\mathcal{G}}}_{\text{outputs}}\right) = A_i F_i^X\left(\underbrace{\{x_{i,jg'}\}_{j\in\mathcal{N},g'\in\mathcal{G}}}_{\text{product } g' \text{ from } j}, L_i, K_i\right)$$

where

$$F_i^Q(\cdot) = \left(\sum \delta_{ig} q_{ig}^{\frac{\sigma+1}{\sigma}}\right)^{\frac{\sigma}{\sigma+1}}, \quad F_i^X(\cdot) = \left(\sum \omega_{i,jg'} x_{i,jg'}^{\frac{\theta-1}{\theta}} + \omega_{i,L} L_i^{\frac{\theta-1}{\theta}} + \omega_{i,K} K_i^{\frac{\theta-1}{\theta}}\right)^{\frac{\theta}{\theta-1}}$$

distance to frontier

Calibration and Results

- Chilean product-level markups, 2018; estimated $\sigma = 1.589$
- Benchmark demand elasticity $\theta = 2.5$ (Arkolakis et al., 2023)

Specification	σ	Welfare loss
Joint production	1.589	20.05%
Independent product lines	$\to \infty$	26.84%

Second-order welfare losses as a percentage of GDP in the horizontal economy.

- Joint production reduces the benchmark loss by about 25%

Conclusion

- Sufficient statistics to measure misallocation with joint production
- Provide empirical evidence consistent with joint production
- Develop and implement distance to frontier formula with joint production
- Ignoring joint production overestimates misallocation

Decreasing Return to Scale in Firm Level

- Production function from Almunia et al. (2021)

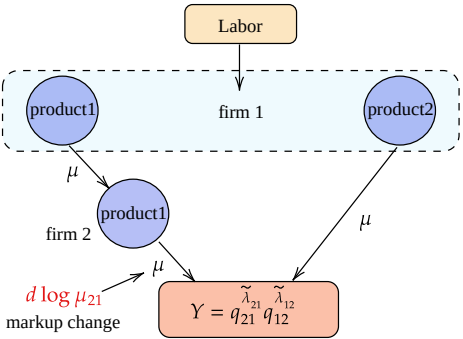
$$q_{11} + q_{12} = L_1^{\alpha}$$

- This is a special case of parametric version of our specification as $\sigma \to \infty$

$$\left(q_{11}^{\frac{\sigma+1}{\sigma}} + q_{12}^{\frac{\sigma+1}{\sigma}} \right)^{\frac{\sigma}{\sigma+1}} = L_1^{\alpha} K^{1-\alpha}$$

where K is firm specific factor

Derivation



$$\left(q_{11}^{\frac{\sigma+1}{\sigma}} + q_{12}^{\frac{\sigma+1}{\sigma}}\right)^{\frac{\sigma}{\sigma+1}} = L$$

- The downstream markup change implies that sales of product 1 of firm 1 changes by $-d \log \mu_{21}$ back
- Sales of product 2 do not change (due to Cobb-Douglas)
- Combining these with $d \log (p_{11}/p_{12}) = \frac{1}{\sigma} d \log (q_{11}/q_{12})$ implies

$$d \log (p_{11}/p_{12}) = -\frac{1}{\sigma+1} d \log \mu_{21}$$

Example with Taste Shock

- Instead of markup shock, consider taste shock: $d\tilde{\lambda}_2 = -d\tilde{\lambda}_1$ back
- Using $d\log(p_{11}/p_{12}) = \frac{1}{\sigma+1}\frac{1}{\tilde{\lambda}_2} d\log\tilde{\lambda}_1$, the TFP response can be expressed by:

$$\begin{aligned}\Delta \log TFP &= -\left(1 - \frac{1}{\sigma+1}\right)\frac{\tilde{\lambda}_1}{\tilde{\lambda}_2}\left(\frac{\bar{\Gamma}_1}{\Gamma_{11}} - 1\right) d\log\tilde{\lambda}_1, \\ &= \underbrace{-\frac{\tilde{\lambda}_1}{\tilde{\lambda}_2}\left(\frac{\bar{\Gamma}_1}{\Gamma_{11}} - 1\right) d\log\tilde{\lambda}_1}_{\text{Single-Product Term}} + \underbrace{\tilde{\lambda}_1\left(\frac{\bar{\Gamma}_1}{\Gamma_{11}} - 1\right) d\log(p_{11}/p_{12})}_{\text{Multi-Product Term}}\end{aligned}$$

- If $\sigma \to \infty$, multi-product term is zero
- If $\sigma \to 0$, $\Delta \log TFP$ is zero

Sufficient Statistics with Taste Shock

Proposition

In the simple example, TFP response to the taste shock is expressed by

$$\Delta \log TFP = \underbrace{Cov_{\boldsymbol{s}_1}\left(d\log p_{(1,\cdot)}, \frac{\bar{\Gamma}_1}{\Gamma_{(1,\cdot)}}\right)}_{\text{Multi-Product Term}} - \underbrace{d\log \Lambda}_{\text{Labor Share}}$$

Single-Product Term

where $d\log p_{(1,\cdot)} = (d\log p_{11}, d\log p_{12})$, $\Gamma_{(1,\cdot)} = (\Gamma_{11}, \Gamma_{12})$, $\bar{\Gamma}_1 = \left(\tilde{\lambda}_{11}\Gamma_{11}^{-1} + \tilde{\lambda}_{12}\Gamma_{12}^{-1}\right)^{-1}$ *and* $\boldsymbol{s}_1 = \left(\tilde{\lambda}_{11}, \tilde{\lambda}_{12}\right)$

Joint Production

Let $J(\boldsymbol{q}, \boldsymbol{x})$ be the joint production function (Hall (1973))

$$J(\boldsymbol{q}, \boldsymbol{x}) = 0,$$

where $\boldsymbol{q}$; output vector and $\boldsymbol{x}$: input vector

Assumptions:

(1) CRS: $J(\boldsymbol{q}, \boldsymbol{x}) = 0$ implies $J(\lambda \boldsymbol{q}, \lambda \boldsymbol{x}) = 0$

(2) Separability between input and output bundles: $J(\boldsymbol{q}, \boldsymbol{x}) = -F^{Q}(\boldsymbol{q}) + F^{X}(\boldsymbol{x})$

Example Constant-Elasticity of Transformation and CES Input (CET-CES): cost function

$$\underbrace{\left(\sum_{g} q_g^{\frac{\sigma+1}{\sigma}}\right)^{\frac{\sigma}{\sigma+1}}}_{\text{Output bundle}} = A \underbrace{\left(\omega_L L^{\frac{\theta-1}{\theta}} + \omega_K K^{\frac{\theta-1}{\theta}}\right)^{\frac{\theta}{\theta-1}}}_{\text{Input Bundle}}$$

back to simple example | back to network settings

Cost Function

Example

CET-CES Cost Function

$$C(\mathbf{q}, w, r; A) = \frac{1}{A}\left(\sum_g \delta^g [q_g]^{\frac{\sigma+1}{\sigma}}\right)^{\frac{\sigma}{\sigma+1}} \left(\alpha_L w^{1-\theta} + \alpha_K r^{1-\theta}\right)^{\frac{1}{1-\theta}},$$

- w and r are (factor) prices
- marginal cost

$$mc_g = \frac{\partial C(\mathbf{q}, w, r; A)}{\partial q_g}$$

back to joint production

Input-Output Objects: Revenue-Based

- To derive sufficient statistics, define Ω to be $(\mathcal{NG}+\mathcal{F})\times(\mathcal{NG}+\mathcal{F})$ input-output matrix: back

$$\Omega_{ig,jg'} = \underbrace{\frac{p_{jg'}x_{i,jg'}}{\mu_{ig}\left(\sum_{j,g'}p_{jg'}x_{i,jg'}+\sum_f w_f L_{if}\right)}}_{\text{Revenue share of } jg'},$$

- Leontief inverse matrix, Ψ:

$$\Psi \equiv (I-\Omega)^{-1},$$

- Domar weights, λ:

$$\begin{aligned}\lambda' \equiv & b'\Psi,\\ = & b' + b'\Omega + b'\Omega^2 + \ldots\end{aligned}$$

λ = sales share (over GDP)

Input-Output Objects: Revenue vs Cost-Based

- Ω and $\tilde{\Omega}$ are input output matrices: back

$$\Omega_{ig,jg'} = \frac{p_{jg'} x_{i,jg'}}{\underbrace{\mu_{ig}\left(\sum_{j,g'} p_{jg'} x_{i,jg'} + \sum_f w_f L_{if}\right)}_{\text{Revenue share of } jg'}}, \qquad \tilde{\Omega}_{ig,jg'} = \frac{p_{jg'} x_{i,jg'}}{\underbrace{\sum_{j,g'} p_{jg'} x_{i,jg'} + \sum_f w_f L_{if}}_{\text{Cost share of } jg'}}$$

- Leontief inverse matrices:

$$\Psi \equiv (I - \Omega)^{-1}, \qquad \tilde{\Psi} \equiv (I - \tilde{\Omega})^{-1}$$

- Domar weights ($\tilde{\Lambda}$ for factors) :

$$\lambda' \equiv b'\Psi, \qquad \tilde{\lambda}' \equiv b'\tilde{\Psi}$$

If there is no markup, $\lambda = \tilde{\lambda}$ = sales share (over GDP)

Cumulative Wedge

- **Cumulative wedge** (for product g of firm i)**:** back

$$\Gamma_{ig} \equiv \underbrace{\tilde{\lambda}_{ig}/\lambda_{ig}}_{\text{downstream wedge}} \times \underbrace{\mu_{ig}}_{\text{own wedge}},$$

- Captures indirect wedges through the downstream supply chain
- In the previous simple example: $\Gamma_{11} = \mu^2 > \Gamma_{12} = \mu$
- Firm-level (cost-based) Domar weights and their shares: $\tilde{\lambda}_i = \sum_g \tilde{\lambda}_{ig}, \quad s_{ig} = \frac{\tilde{\lambda}_{ig}}{\tilde{\lambda}_i}$

Decomposition of TFP into Allocative Efficiency and Technology

- $\mathcal{Y}(X, A)$: output Y given firm level productivities A and shares $X_{ijg} = x_{ijg}/q_{jg}$
- Change in response to shocks:

$$d \log Y = \underbrace{\frac{\partial \log \mathcal{Y}}{\partial \log X} d \log X}_{\Delta \text{Allocative Efficiency}} + \underbrace{\frac{\partial \log \mathcal{Y}}{\partial \log A} d \log A}_{\Delta \text{Technology}}$$

Decomposition of TFP into Allocative Efficiency and Technology

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- Efficient economy (set all markups equal 1) implies Hulten (1978):

$$\underbrace{d\log Y - \sum_f \tilde{\Lambda}_f d\log F_f}_{d\log TFP} = \underbrace{0}_{\Delta\text{Allocative Efficiency}} + \underbrace{\sum_i \tilde{\lambda}_i d\log A_i}_{\Delta\text{Technology}}$$

back

Allocative Efficiency: Multi-Product Term

$$\text{Multi-Product Term} = \sum_i \tilde{\lambda}_i Cov_{\mathbf{s}_i}\left(\underbrace{d \log p_{(i,\cdot)}}_{\text{Price changes}}, \underbrace{\frac{\bar{\Gamma}_i}{\Gamma_{(i,\cdot)}}}_{\text{Normalized inverse wedges}} \right)$$

- Inside firm:
  - Joint production $\Rightarrow$ the opportunity cost to modify firm i's product mix is *not* constant
  - Larger price increases for low-Γ_{ig} products $\Rightarrow$ a positive multiproduct term
  - In Chile, this correction offsets part of the negative single-product term
- Aggregation:
  - Sum up the covariance using firm's Domar weights, $\tilde{\lambda}_i$ back

Allocative Efficiency: Single-Product Term

$$\text{Single-Product Term} = -\sum_f \underbrace{\tilde{\Lambda}_f d\log\Lambda_f}_{\text{(1) Aggregate Factor share}} - \sum_i \underbrace{\tilde{\lambda}_i d\log\mu_i}_{\text{(2) Firm-level Markup}}$$

(1) A decline in factor shares $\Rightarrow$ resources shifting to high markup firms that are underproducing

(2) The direct change in factor shares due to markup changes: To isolate changes in allocative efficiency, need to subtract this back

Efficient Benchmark

- Efficient economy (set all markups equal 1) implies Hulten (1978):

$$d\log TFP = \underbrace{0}_{\Delta\text{Allocative Efficiency}} + \underbrace{\sum_i \tilde{\lambda}_i d\log A_i}_{\Delta\text{Technology}}$$

back

Empirical Evidence of Joint Production and Parametric Counterfactual

- Product-specific demand shocks reveal technological links between products within firms
- A CET production function summarizes product-mix flexibility by the transformation elasticity σ
- Estimate σ using the same demand-shock variation
- Use this estimate for product-level markups and an illustrative distance-to-frontier exercise

Motivating Empirical Evidence of Joint Production

Assume:

- The cost function is $C(q_1, \ldots, q_G) = P_I \left(\sum_{g=1}^{G} \omega_g q_g^{\frac{\sigma+1}{\sigma}} \right)^{\frac{\sigma}{\sigma+1}}$
- Demand for each product is $q_g = D_g p_g^{-\theta_g}$
- Each product's price maximizes profits given $\{D_g\}$ and input prices variable markup

Proposition

A negative demand shock to product m ($d \log D_m < 0$) lowers quantity and raises the price of every other product $g \neq m$ if and only if $\sigma < \infty$.

- Lower production of m raises the marginal cost of other jointly produced goods

Data Sources

- Universe of electronic invoices between formal Chilean firms, aggregated to firm–product–month
- Monthly quantities and prices, January 2019–December 2021
- March 2020 lockdowns affecting buyers provide product-specific demand shocks buyer regression



Locked-down municipalities, March 2020

Simple Event Study Specification

- Main product m: highest revenue for firm i over 2019–2021
- Treated if at least one buyer of m is in a municipality locked down in March 2020
- Estimate spillovers to other products $g \neq m$:

$$\log X_{igt} = \sum_{j=-11}^{10} \beta_j Lockdown_{i,t-j} + FE_{ig} + FE_t + \varepsilon_{igt}$$

- X: quantity or price; FE_{ig}: firm–product effects; FE_t: month effects
- $Lockdown_{i,t-j} = 1$ if the main-product demand shock occurred j months ago

Sample Restriction

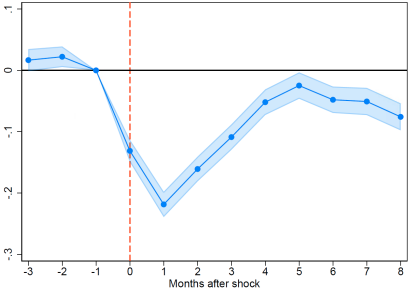
- Isolate demand shocks to the main product:
  - (1) The firm and its suppliers are outside locked-down areas
  - (2) Buyers of non-main product g are outside locked-down areas
- Exclude sellers below the median transaction value, retaining over 99% of transaction value
- Compare firms whose main-product buyers were affected with firms whose buyers were unaffected sample

Results

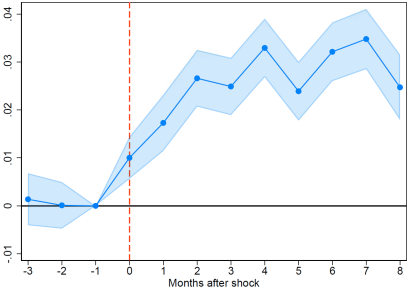
Negative demand shocks to the main product reduce quantities and raise prices of other products.

robustness

Log quantity



Log price



95% confidence intervals; firm–municipality clustering. Month 0 is March 2020; displayed window: -3 to $+8$ months.

Estimating Elasticity of Transformation σ

$$\Delta \log\left(\frac{p_{ig}}{p_{im}}\right) = \alpha + \beta \Delta \log\left(\frac{q_{ig}}{q_{im}}\right) + \text{Fixed effects} + \xi_{igt}, \qquad \beta = 1/\sigma$$

Yearly differences; instrument relative quantities with main-product buyers' lockdown exposure.

	(1)	(2)	(3)
β	0.865***	1.207***	0.630***
	(0.063)	(0.259)	(0.137)
Implied σ	1.156	0.828	1.589
Time FE	N	Y	Y
Industry FE	N	N	Y
First-stage F	699.22	52.00	98.56

All columns: 2SLS; standard errors clustered at the firm–municipality level. *** $p < 0.01$.

Estimates reject independent product lines ($\sigma \to \infty$). back to measurement

Buyer-Level Demand Shock

$$\log M_{it} = \alpha Lockdown_{it} + FE_t + FE_i + \varepsilon_{it}, \qquad M_{it}: \text{ total intermediate purchases.}$$

	(1)	(2)	(3)
Lockdown	-0.222^{***}	-0.230^{***}	-0.191^{***}
	(0.0524)	(0.00521)	(0.0589)
Firm FE	Y	Y	Y
Time FE	N	N	Y
Sector $\times$ Time FE	N	Y	N
No suppliers in lockdown areas	N	N	Y

January 2019–March 2020. OLS; standard errors clustered at the firm–municipality level. $^{***}p < 0.01$.

Purchases by buyers in lockdown municipalities fell by about 20%. back

Characteristics of Treated and Control Firms

	Treated	Control
Number of firms	1,432	155,247
Workers	3	3
Products sold	16	10
Providers	98	134
Buyers	24	8
Annual revenue (million pesos)	412	319
Annual intermediate purchases (million pesos)	107	59

February 2020 characteristics. Treated firms have main-product buyers subject to lockdowns in March 2020. All rows except the number of firms report medians.

back

DiD Robustness: Responses of Other Products

$$\log X_{igt} = \beta \, Lockdown_{it} + FE_{ig} + FE_t + \varepsilon_{igt}.$$

	Baseline	Input price control	Large firms	Manufacturing	Product × Time FE	Continuous treatment
$\log q$	-0.117^{***}	-0.117^{***}	-0.102^{***}	-0.106^{***}	-0.125^{***}	-0.159^{***}
	(0.0046)	(0.0046)	(0.0053)	(0.0188)	(0.0049)	(0.0062)
$\log p$	0.013^{***}	0.013^{***}	0.015^{***}	0.014	0.012^{***}	0.013^{***}
	(0.0022)	(0.0022)	(0.0024)	(0.0096)	(0.0023)	(0.0024)
Observations	7,693,066	7,669,848	4,394,166	1,399,617	7,594,918	7,693,066

All specifications include firm-product and time FE. Columns 2–6 control for input prices. Large firms are above the 80th sales percentile. Continuous treatment is the main product's sales share to lockdown destinations. $^{***}p < 0.01$.

Quantities fall across specifications; the manufacturing price estimate is imprecise. back

Variable Markup

- If Marshall's second law of demand holds (the higher q, the higher the markup, the smaller the pass-through), the same proposition holds
- Intuitively, markups offset some of the price changes back

Proposition

Under Marshall's second law of demand, a negative demand shock to product k ($d \log D_k < 0$) affects product $g \neq k$:

(i) If $\sigma < \infty$, quantity: $d \log q_g < 0$, price: $d \log p_g > 0$,

(ii) If $\sigma = \infty$, quantity: $d \log q_g = 0$, price: $d \log p_g = 0$

Data and Growth Accounting

Growth-accounting sample: 2016–2022.

(1) Sales, materials, investment: F29.

(2) Wage bill and employment: DJ1887 (coverage: 2005–2022).

(3) Capital stock: F22, combined with investment using perpetual inventory methods.

- Rental rate: 10-year government bond rate minus expected inflation, plus depreciation (LA-Klems) and an external financing premium.

(4) Products, I–O matrices, and output and input prices: firm-to-firm electronic invoices (2016–2022).

(5) Official deflators for aggregate real variables.

back

Data Cleaning

- Require positive sales, materials, wage bill, and capital.
- Exclude firms with fewer than two employees or capital below US$20.
- Winsorize variables at the 1st and 99th percentiles.

Multi-Product Firms Dominate Intermediate Transactions

Multi-product firms account for 99% of intermediate transaction value.

Percentile	Products per firm
1%	1
5%	2
10%	4
25%	36
50%	**475**
75%	2,459
90%	32,195
95%	37,422
99%	62,372

Weighted by transaction value, 2018

back unweighted

Unweighted Distribution of the Number of Products

Percentile	Number of products by firm
1%	1
5%	1
10%	1
25%	2
50%	**7**
75%	26
90%	119
95%	290
99%	1,253

back

Baseline: Product-Level Markup Estimation

- Extend Dhyne et al. (2022) to CET output aggregation with $\sigma = 1.589$ and common productivity across a firm's products.
- Estimate the input-bundle technology by GMM within one-digit product categories:

$$\log Q_{it} = \beta_0 + \beta_K \log K_{it} + \beta_L \log L_{it} + \beta_M \log M_{it} + \omega_{it}.$$

- Törnqvist price indexes recover quantity-based elasticities; lagged inputs serve as instruments.

$$\mu_{ig} = \frac{\beta_M}{\eta^Q_{ig}} \frac{R_{ig}}{P^M_i M_i}, \qquad \eta^Q_{ig} \equiv \frac{\partial \log F^Q_i}{\partial \log q_{ig}} = \frac{q_{ig}^{(\sigma+1)/\sigma}}{\sum_{g' \in \mathcal{G}_i} q_{ig'}^{(\sigma+1)/\sigma}}.$$

β_M: materials elasticity of the common input bundle. η^Q_{ig}: elasticity of the output aggregator with respect to product g. R_{ig}: product revenue; $P^M_i M_i$: materials expenditure.

back

Most Cumulative-Wedge Variation Is Downstream

$$\log \Gamma_{ig} = \log(\tilde{\lambda}_{ig}/\lambda_{ig}) + \log \mu_{ig}.$$

Year	Downstream wedge	Own markup	Covariance
2016	92.4%	7.3%	0.3%
2017	92.4%	7.4%	0.2%
2018	92.5%	6.9%	0.6%
2019	92.9%	6.6%	0.5%
2020	92.9%	6.2%	0.9%
2021	93.2%	5.4%	1.4%
2022	93.9%	4.9%	1.2%

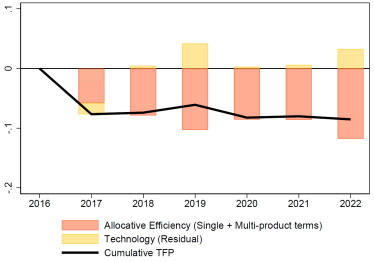
Shares of $\text{Var}(\log \Gamma_{ig})$; the covariance column includes the factor of two.

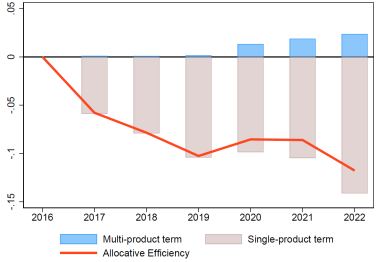
Downstream wedges account for more than 92% of the variation. back

Accounting Markups in the Multiproduct Term

Hold the single-product term fixed; replace own markups only in the multiproduct term.

$$\Gamma_{ig}^{acct} = \frac{\tilde{\lambda}_{ig}}{\lambda_{ig}} \mu_i^{acct}, \qquad \mu_i^{acct} = \frac{\text{firm sales}}{\text{firm costs}}.$$



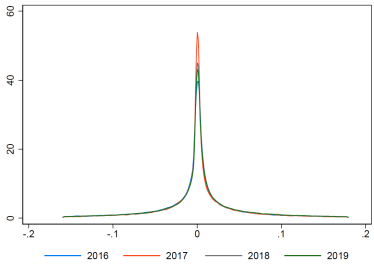


The AE decline is 11.77%; ignoring joint production still overstates it by about 19%. More than half of the baseline correction remains. back

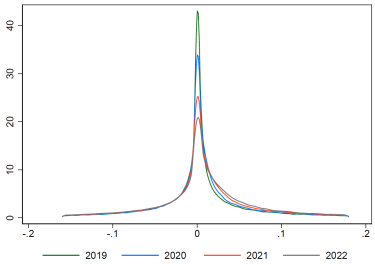
Distribution of the Firm-Level Multiproduct Term

$$\mathrm{Cov}_{\boldsymbol{s}_i}\left(d\log p_{(i,\cdot)}, \frac{\bar{\Gamma}_i}{\Gamma_{(i,\cdot)}}\right)$$

2016–2019



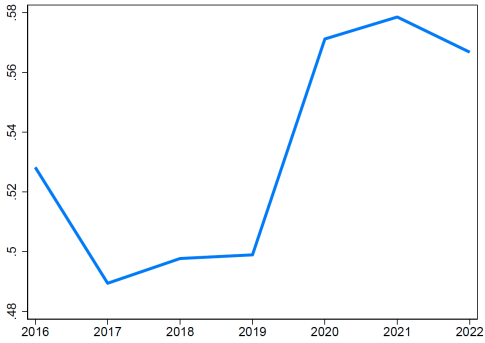
2019–2022



The distribution is stable before COVID and shifts to the right afterward. product mix adjustments back

Large Product Mix Adjustments: Suggestive Evidence

The median within-firm variance of product quantity changes rises after 2019.



Median across firms of $\mathrm{Var}_{\lambda_i}(d \log q_{ig})$. This coincides with the larger multiproduct correction during COVID. back

Distance to the Pareto-Efficient Frontier with Joint Production

Proposition

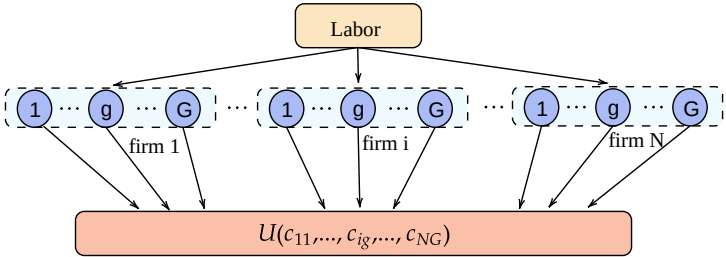
Starting at an efficient equilibrium, the second-order TFP loss from introducing wedges is given by Domar-weighted Harberger triangles:

$$\mathcal{L} \equiv -\Delta \log TFP = -\frac{1}{2} \sum_{ig} \lambda_{ig} \underbrace{d \log q_{ig}}_{\text{quantity}} \underbrace{d \log \mu_{ig}}_{\text{markup}}$$

where λ_{ig} is the Domar weight of product g from firm i back

Horizontal Economy

- A representative consumer (CES with elasticity θ)
- N multi-product firms use labor to make G products using CET technology (elasticity σ)
- Markups μ_{ig} are heterogeneous across products and firms



Analytical Structural Result: Horizontal Economy

Starting from an efficient equilibrium, the positive welfare loss from introducing markup distortions is, to second order:

$$\mathrm{L} \equiv -\Delta \log \mathrm{TFP} \approx \frac{\theta}{2}\left[\underbrace{\mathrm{Var}_{\bar{\lambda}}\left(d\log\mu_i\right)}_{\text{between firms}} + \frac{\sigma}{\sigma+\theta}\underbrace{E_{\bar{\lambda}}\left\{\mathrm{Var}_{s_i}\left(d\log\mu_{ig}\right)\right\}}_{\text{within firms}}\right].$$

$$\lambda_i = \sum_g \lambda_{ig}, \qquad s_{ig} = \frac{\lambda_{ig}}{\lambda_i}, \qquad d\log\mu_i = E_{s_i}[d\log\mu_{ig}].$$

Here $\bar{\lambda} = (\lambda_1, \ldots, \lambda_N)$ gives firm-level Domar weights.

- Joint production attenuates the within-firm component by limiting product-mix adjustment.
- $\sigma \to \infty$: independent product lines; all markup dispersion matters.
- $\sigma \to 0$: only between-firm dispersion in average markup changes matters.

back to environment calibration and results

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